add much sugar, you will reach a stage at which the added sugar remains as solid and the resulting solution is called a saturated solution. Here, as in the previous cases a dynamic equilibrium is established between the solute molecules in the solid phase and in the solution phase.

$$\text{Sugar (Solid)} \rightleftharpoons \text{Sugar (Solution)}$$

In this process

$$\text{Rate of dissolution of solute} = \text{Rate of crystallisation of solute}$$

Gas in liquids

When a gas dissolves in a liquid under a given pressure, there will be an equilibrium between gas molecules in the gaseous state and those dissolved in the liquid.

Example:

In carbonated beverages the following equilibrium exists.

$$CO_2\,(g) \rightleftharpoons CO_2\,(\text{Solution})$$

Henry's law is used to explain such gas-solution equilibrium processes.

8.3. Chemical Equilibrium

Similar to physical processes chemical reactions gradually attain a state of equilibrium after sometime. Let us consider a general reversible reaction.

$$A + B \rightleftharpoons C + D$$

Initially only A and B are present. Soon, a few molecules of the products C and D are formed by the forward reaction. As the concentration of the products increases, more products collide and react in the backward direction. This leads to an increase in the rate of backward reaction. As the rate of reverse reaction increases, the rate of the forward reaction decreases. Eventually, the rate of both reactions becomes equal.

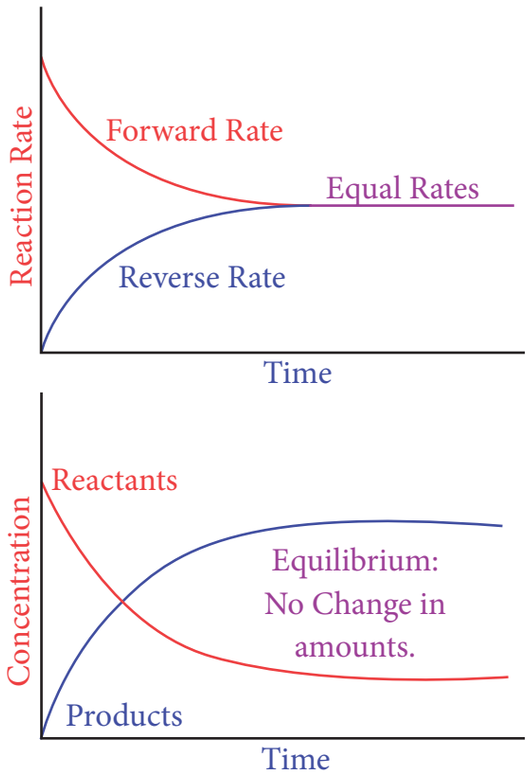


Figure: 8.3 Attainment of equilibrium state for the variation of the rates of forward and backward reactions with time.

8.4. Dynamic nature of equilibrium:

Let us consider, a situation in multi storey building, people are moving from first floor to second floor and vice versa. Assume that a certain number people moves up from first floor to second floor in a specific time, and the same number of people moves down from second floor to the first floor in the same time. Now, the rate of movement of people from first to second floor equals the

11th Chemistry Chapter-8 Vol-2 EM.indd 4 09-01-2023 14:38:22